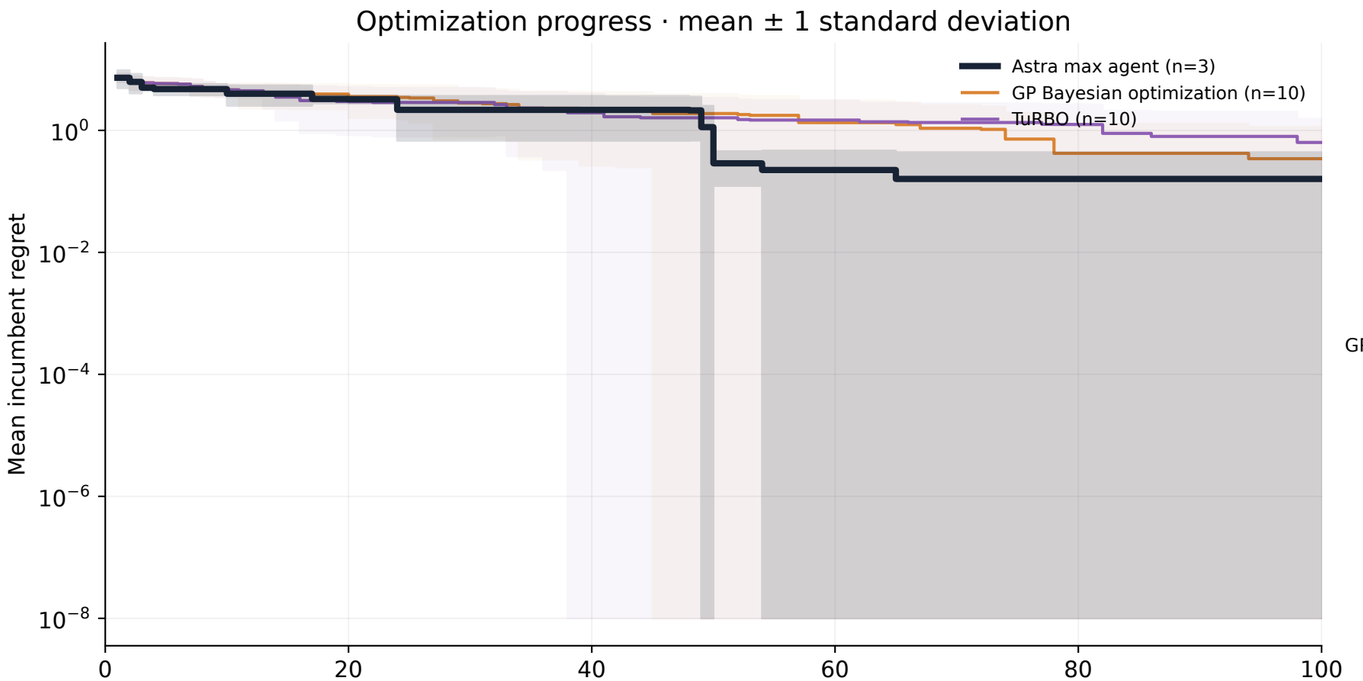
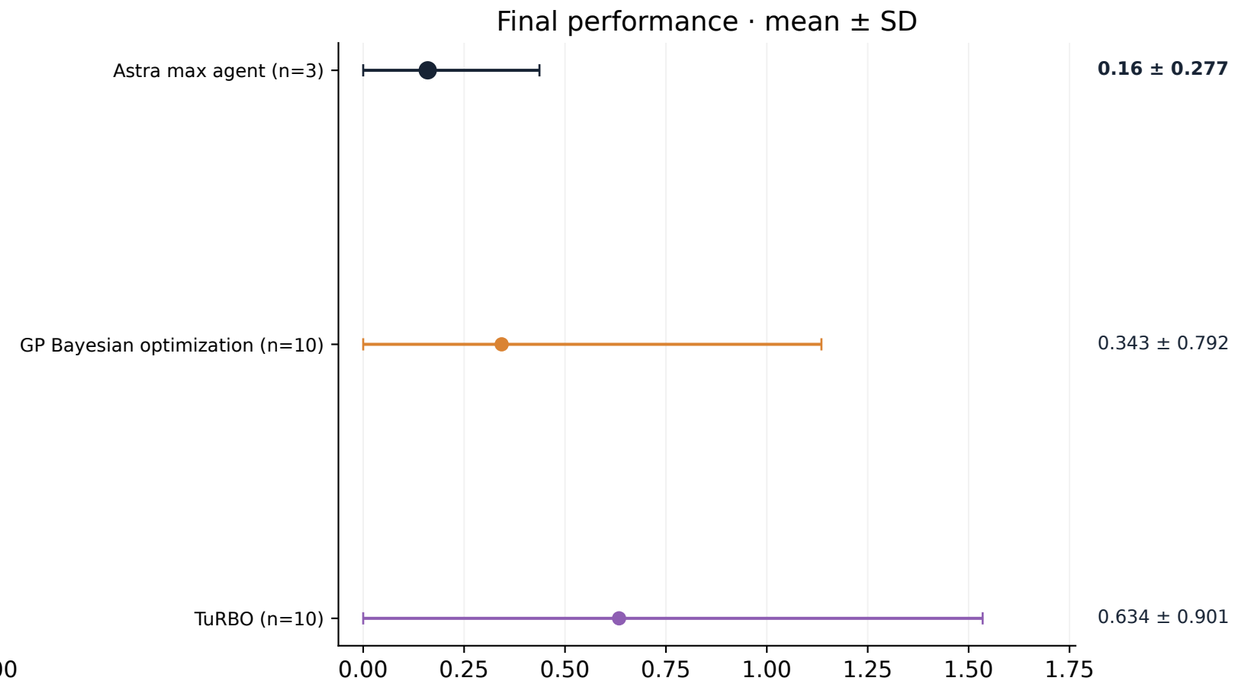
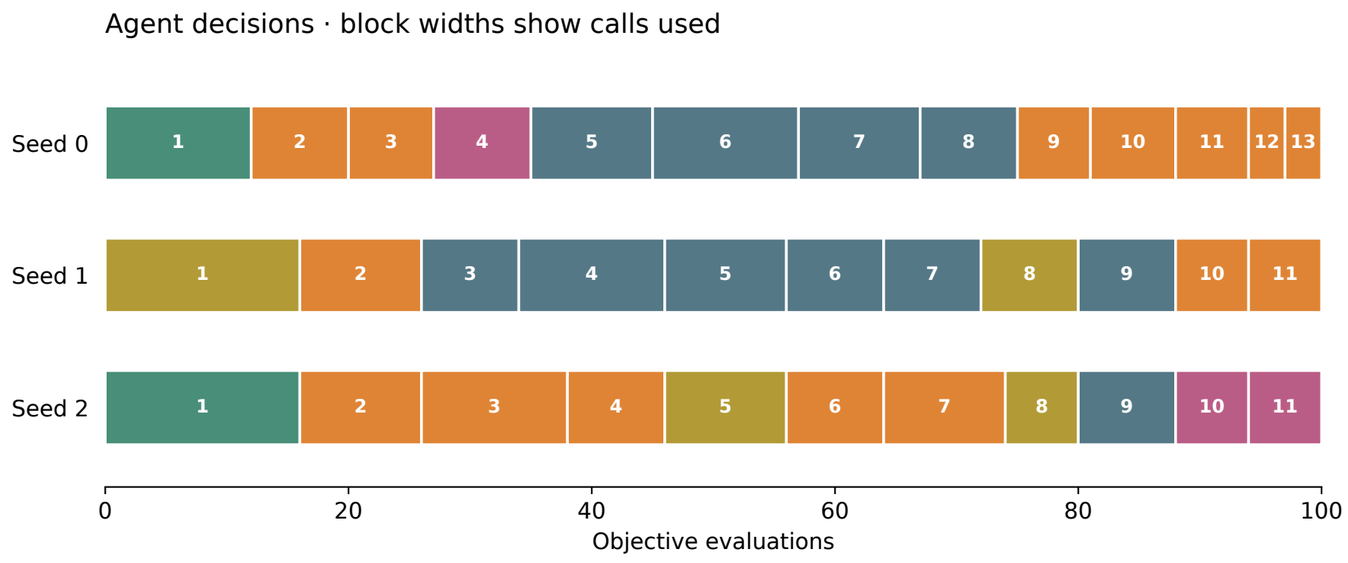
MOF candidate pool — autonomous agent versus standalone optimizers

1,000 candidates · 193 features · GPT-6 Astra, max reasoning · 3 agent seeds · 100 total evaluations per run







Seed 0
1. Diverse design (12) · 2. GP-BO (8) · 3. GP-BO (7) · 4. GP-BO (8) · 5. Tree model (10) · 6. Tree model (12) · 7. Tree model (10) · 8. Tree model (8) · 9. GP-BO (6) · 10. GP-BO (7) · 11. GP + trees (6) · 12. GP + trees (3) · 13. GP + trees (3)

Seed 1
1. Diverse design (16) · 2. GP-BO (10) · 3. Tree model (8) · 4. Tree model (12) · 5. Tree model (10) · 6. Tree model (8) · 7. Tree model (8) · 8. Tree model (8) · 9. Tree model (8) · 10. GP + trees (6) · 11. GP + trees (6)

Seed 2
1. Cluster design (16) · 2. GP-BO (10) · 3. GP-BO (12) · 4. GP-BO (8) · 5. Tree model (10) · 6. GP-BO (8) · 7. GP-BO (10) · 8. Local check (6) · 9. sparse ensemble ei (8) · 10. sparse ensemble lcb (6) · 11. sparse ensemble lcb (6)

Curves and markers: arithmetic means. Shading/error bars: sample standard deviation (ddof=1), not confidence intervals; lower limits clipped at zero.
Agent n=3; each baseline n=10. MOF baselines use calls 1–100 of the supplied 110-call traces, including all ten initialization calls. Regret = full-pool maximum − best observed score.
Decision boundaries are prospective evaluator snapshots; parentheses give block call counts. Exact labels, settings and rationale are preserved in the per-seed reports.
Log display only: zero values and clipped band endpoints are drawn at 1e-08.